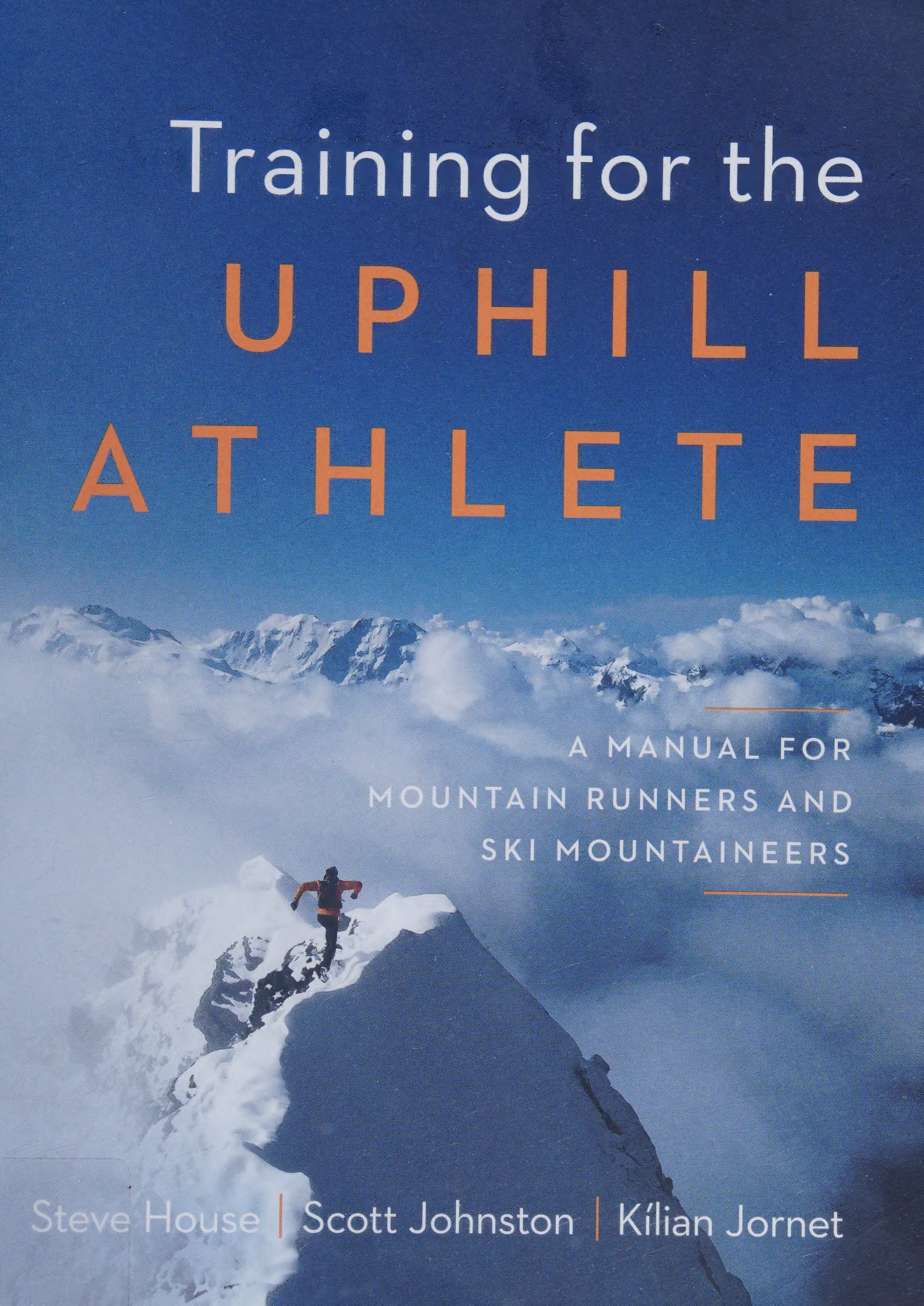
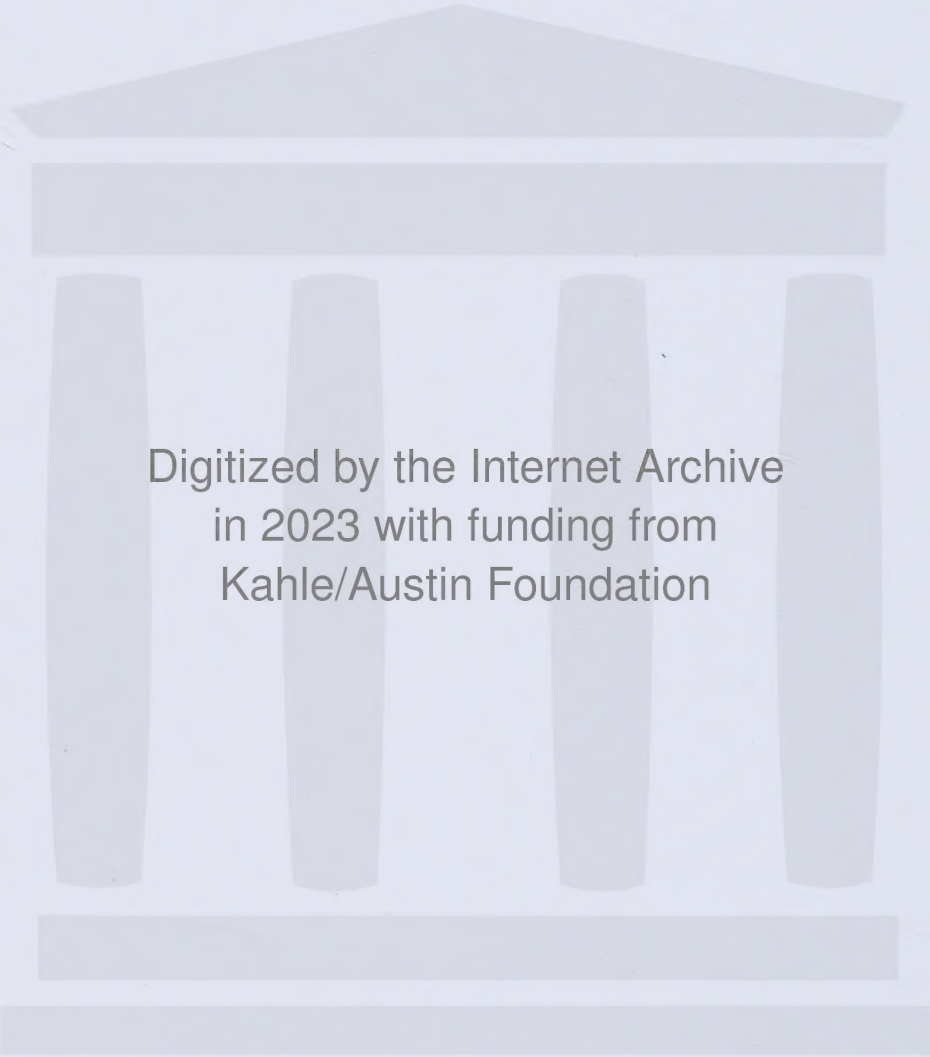


The background of the entire cover is a photograph of a person in a red jacket running along the ridge of a snow-covered mountain peak. The sky is a deep blue, and there are white clouds below the mountain range.

Training for the UPHILL ATHLETE

A MANUAL FOR
MOUNTAIN RUNNERS AND
SKI MOUNTAINEERS

Steve House | Scott Johnston | Kilian Jornet



Digitized by the Internet Archive
in 2023 with funding from
Kahle/Austin Foundation

<https://archive.org/details/trainingforphil0000jorn>

Training for the
U P H I L L
A T H L E T E

Training for the UPHILL ATHLETE

A MANUAL
FOR MOUNTAIN
RUNNERS AND SKI
MOUNTAINEERS

Kilian Jornet | Steve House | Scott Johnston

patagonia

Training for the Uphill Athlete
A Manual for Mountain Runners and Ski Mountaineers

Patagonia publishes a select number of titles on wilderness, wildlife, and outdoor sports that inspire and restore a connection to the natural world.

copyright 2019 Patagonia Works
Text © Steve House, Kilian Jornet, Scott Johnston
Graphs, tables, and illustrations © Patagonia

Photograph copyrights held by the photographer indicated in captions.

All rights reserved. No part of this book may be used or reproduced in any manner whatsoever without written permission from the publisher and copyright holders. Requests should be emailed to books@patagonia.com or mailed to Patagonia Books, Patagonia Inc., 259 W. Santa Clara St., Ventura, CA 93001-2717.

Disclaimer: The information provided in this book is designed to provide helpful information on the subjects covered, including endurance and strength training, nutrition, and physical and mental fitness. It is particularly designed for people who are actively participating in or would like to train for certain extreme physical activities such as rock and ice climbing. This information is not for everyone, as many of the activities described are intense and can be dangerous. You should not undertake any of these activities unless and until you have consulted with qualified medical personnel and been cleared to participate. And, of course, if you find that you are suffering from the symptoms of any physical or medical condition, you should consult with your physician or other qualified health provider immediately. Do not avoid or disregard professional medical advice or delay in seeking it because of something you have read in this book.

All of the information contained in this book, including text, graphics, images, recipes, exercises, training regimens, diets, and references to third party materials are for informational and educational purposes only. The authors and publisher of this book shall have no liability or responsibility to any reader or any third party arising out of any injury or damage incurred as a result of the use of the information provided in this book.

Hardcover edition
Printed in Canada on 100 percent post-consumer recycled paper

Editor - John Dutton
Photo Editor - Kyle Sparks
Book Designer - Faceout Studio
Project Manager - Jennifer Patrick
Photo Production - Sus Corez
Production - Rafael Dunn
Creative Directors - Christina Speed, Bill Boland
Creative Advisor - Jennifer Ridgeway
Director of Books - Karla Olson



ENVIRONMENTAL BENEFITS STATEMENT				
Patagonia Inc saved the following resources by printing the pages of this book on chlorine free paper made with 100% post-consumer waste.				
TREES	WATER	ENERGY	SOLID WASTE	GREENHOUSE GASES
319	26,000	134	1,100	138,000
FULLY GROWN	GALLONS	MILLION BTUS	POUNDS	POUNDS

 Environmental impact estimates were made using the Environmental Paper Network Paper Calculator 4.0. For more information visit www.papercalculator.org.

Cover photos: (front) Kilian Jornet in Switzerland. Photo: Steve House
(back) Kim Strom runs a rocky ridgeline above Saas Almagell, Switzerland. Photo: Dan Patitucci
Paperback 978-1-938340-84-0
E-Book 978-1-938340-85-7
Library of Congress Control Number 2018965947

CONTENTS

FOREWORD

Nose to the Wind..... 11

CHAPTER ONE

How to Use This Book..... 17

SECTION ONE

The Physiological Basis of Endurance Training..... 21

CHAPTER TWO

The Physiology of Endurance..... 23

- The Evolution of Endurance 23
- Endurance and Fatigue 25
- The Factors That Determine Endurance Performance 27
- ATHLETE STORIES** Making the Decision – Janelle Smiley..... 29
- Endurance Seen Through the Metabolic Lens 33
- The Implications for Performance 45
- ATHLETE STORIES** Nickels and Dimes – Rod Bien 51
- Fat Adaptation and Endurance Performance 55
- Enhancing Your Aerobic and Anaerobic Metabolic Pathways 64
- ATHLETE STORIES** The Roots of My Athleticism – Kilian Jornet 66

SECTION TWO

The Methodological Basis of Endurance Training	69
--	----

CHAPTER THREE

The Methodologies of Endurance Training	71
A Little History	71
Important Concepts, Terms, and Principles	72
Understanding the Language of Intensity	79
ATHLETE STORIES A Long Road to the Same Place I Started - Dakota Jones ...	82
ATHLETE STORIES Training for Mountain Running Without Mountains or Time - John Kelly	86
ATHLETE STORIES The Message of Running - Rickey Gates	93
ATHLETE STORIES In the Mountains, Durability Equals Speed - Mike Foote ...	104
ATHLETE STORIES The Everest Marathon: My Wild Introduction to Ultrarunning - Anna Frost	114

CHAPTER FOUR

Monitoring Your Training	117
Ways to Monitor for Recovery	117
Use a Training Log to Monitor Training Load	126
Returning to Training After a Break	129
ATHLETE STORIES Reframing Adversity - Jared Campbell	134
Overtraining Syndrome (OTS)	136
ATHLETE STORIES A Lesson Learned the Hard Way: Training Isn't Competition - Marc Pinsach Rubirola	138
Periodization	144
ATHLETE STORIES The Life Cycle of an Ultrarunner - Krissy Moehl	147

CHAPTER FIVE

The Application Process: Where Theory Meets Reality	151
Testing	152
Aerobic Base Development: Volume Is the Key	156

Endurance Training Methods	161
ATHLETE STORIES Tracing My Own Lines - Luke Nelson.....	164
ATHLETE STORIES Mike Foote Sets New World Record for Uphill Vertical Skied in Twenty-Four Hours - Laura Larson	186

SECTION THREE

Strength and the Uphill Athlete	191
---------------------------------------	-----

CHAPTER SIX

Strength Training for the Uphill Athlete.....	193
General Versus Specific Strength	194
Economy and Muscular Endurance	197
Speed and Strength	198
Muscular Endurance and Strength	198
Summary.....	199
ATHLETE STORIES Don't Forget to Pack the <i>Why</i> - Clare Gallagher.....	200

CHAPTER SEVEN

General Strength Assessment and Improvement.....	203
Assessment and Basic Exercise Programs for Legs and Hips	204
Stage 1 Leg and Hip Program.....	210
Stage 2 Leg and Hip Program.....	211
Stage 3 Leg and Hip Program	214
Assessment and Basic Exercise Programs for Upper Body	214
Stage 1 Upper-Body Program.....	217
Stage 2 Upper-Body Program.....	219
Stage 3 Upper-Body Program.....	221
Core Strength	224
Core Strength Workouts.....	231
Planning Your General Strength Workouts Stages 1 Through 3	234
ATHLETE STORIES It Will Get Better - Anton Krupicka	236

CHAPTER EIGHT

Specific Strength-Training Methods	239
Strength	239
Summary	254

SECTION FOUR

How to Train	257
--------------------	-----

CHAPTER NINE

Programming	259
Beginning the Plan	261
Outline for Planning	262

CHAPTER TEN

Transition Period Training	269
Where to Start	269
General Guidelines for Your Transition Period Planning	272
ATHLETE STORIES How to Plan Your Race Season - Luke Nelson	274

CHAPTER ELEVEN

Introduction to the Base Period (for Both Running and Ski Mountaineering)	277
The Low-Hanging Fruit	278
The Cornerstone of the Base	280
The Mechanics of Planning	281
ATHLETE STORIES Lining Up and Knowing I Would Win - Rico Elmer	283
Base Period Guidelines	302
ATHLETE STORIES The Ghost on the Mountain - Luke Nelson	309

CHAPTER TWELVE

Special Considerations for Skimo and Ski Mountaineering 311

Structuring a Category 1 Athlete’s Skimo and Ski Mountaineering Plan 312

ATHLETE STORIES *Specific Training for a Specific Goal*

 – Javier Martín de Villa 315

Structuring a Category 2 Athlete’s Skimo and Ski Mountaineering Plan..... 321

Training During Race Season for Category 1 and 2 Athletes 326

CHAPTER THIRTEEN

Special Considerations for Mountain Running 337

Volume 337

ATHLETE STORIES Traversing the Borders of Pain - Emma Roca 340

Structuring a Category 1 Athlete’s Mountain Running Training Plan 342

Structuring a Category 2 Athlete’s Mountain Running Training Plan 350

ATHLETE STORIES Wildlife Rendezvous – Jeff Browning 355

ATHLETE STORIES The Happiness Factor – Emelie Forsberg 361

Glossary 363

Index 369



FOREWORD

Nose to the Wind

The wind had made its presence known all night, the tiny tent shaking off its layer of breath-turned-frost with each gust. The icy flakes settled, then melted onto the only part of me sticking out of the sleeping bag: my nose. I was tired when I woke, but no more than Marko, whom I followed out the tent door at dawn. We snapped on crampons, shouldered our summit packs, and climbed into the storm.

All our known world was below us, invisible but pulling on us, like gravity. Marko climbed steadily, quickly. I, more slowly and erratically. Soon he was also invisible to me, somewhere above me. The loneliness of being lost in between my partner and my world was crushing. I sped up for a moment then stumbled, gasping, coughing, and wheezing. With only footprints in the snow—no one had ever passed here before (or since)—and the occasional parting of cloud to reveal Marko's green jacket, the mark of an alien in the black-and-white world of 23,000 feet in the Himalaya, I could see enough to know he was there. I could follow, but I couldn't keep up.

Two hours later we turned around, our summit attempt abandoned after five hard-fought days of climbing. A lifetime goal became the newest addition to the dusty, cluttered shelf of abandoned dreams.

I'd climbed thousands of mountains and I'd never trained. I'd earned sponsorships and attention and money as a climber. And I'd never trained. I'd outpaced partners, put up first ascents, gone sixty hours up the steepest, iciest wall in North America. And I'd never trained. Now I had to turn around on the biggest, highest, hardest mountain I'd ever attempted—because I had never trained.

When I came home I found a coach and I started training. Six months later I was sick, IVs in my veins, fever on my brow. I didn't eat for twenty-one days

and went from 165 pounds to 135 pounds. Once I started to come around, Scott Johnston, my friend, a former World Cup cross-country ski racer, and climbing partner, called me over to his house and cooked a hamburger.

Sitting it in front of me, he said, “You know why you got sick, right? She [your coach] doesn’t understand climbing; she doesn’t understand what goes into your job. She overtrained you. Badly. You have to layer training volume on gradually.”

So began my first of thousands of lessons on endurance training and physiology, lessons that eventually would become our first book together, *Training for the New Alpinism*. Needless to say, Scott became my coach. My fitness blossomed. Less than two years later, I turned pro climber—no other work. My climbing career went world-class, and I never found myself overtrained, ill, or injured again. Until I fell 100 feet from a mountain in Canada.

I like to say that the only people who are not enthusiastic about training are those who haven’t trained correctly. The only people who hate lunges are those who’ve never done enough of them to experience what strong legs feel like on the trail. The only ones who hate long, slow aerobic capacity-building runs and skis are those who have never known what it feels like to sail up the mountain, nose to the wind, with ease. Relaxed, poised, moving fast and flying.

The fitness world is a minefield of snake oil salesmen, fads, and claims that are too good to be true that are just that. While I recovered from my fall, healing dozens of fractures, punctured lungs, and a shaken identity, Scott and I began to try to answer the most common question I got as a professional climber: How do you train? When we published *Training for the New Alpinism* in 2014, its success far exceeded our expectations. We’d figured on enough alpinists to sell maybe 1,000 copies. We sold 50,000.

Eventually we learned that we had been right. Fifty thousand alpinists did not buy our book. Ten thousand did, with the other 40,000 copies going to mountain runners, ski mountaineers, and skimo racers who lacked a training guide of their own and figured alpinism was the closest thing. They bought in, literally and figuratively, to the no-nonsense, no-fads, no-promises approach of *Training for the New Alpinism*.

© 2015 STEVE HOUSE AND SCOTT JOHNSTON

I met Kilian Jornet in my home mountains in early winter of 2014. He and Emelie Forsberg were there to ski some of the famous Colorado powder. We teamed up and I guided them to a classic steep descent near Telluride—the San Joaquin Couloir. We skinned from the valley at the first December light, scrambled up the shoulder, and were back in the valley before noon. At the bakery, we ate muffins, our first food of the day, and Kilian told me how much he’d enjoyed *Training for the New Alpinism*. I told him how surprised we’d been by the response, and by the adoption of the book by so many mountain athletes.



Steve House and Kilian Jornet on top of the Täschhorn, Switzerland. Photo: Steve House Collection

“Yes,” he said. “There’s nothing like it for us.”

“We should write that book together!” I almost shouted. Kílian had told me about his university studies in exercise physiology earlier that morning while we were skinning.

Kílian demurred. “I don’t want to write,” he said, then excused himself to do another workout, steep Zone 3 intervals.

On the drive home I called Scott and a plan was born. At the 14,000-foot camp on Denali that June, I shared our progress with Kílian. That August, with an outline, a publishing agreement, and a lot of emails, Kílian and I met again, this time in Zermatt, Switzerland. We squeezed in some peak bagging—nothing technical. Fortunately for me, fresh snow blanketed the Alps that August; I had the impossible task of trying to keep up with Kílian on steep trails and ridges. We spent three rainy days camped in my van, frying up big pans of rösti (Swiss potatoes and cheese) and discussing the book, training techniques and their relative value, training mistakes, how to organize and present so much information, and more. Our only problem was parking the van someplace acceptable to the Swiss.

Armed with Kílian’s insights, Scott wrote, and wrote, and shared drafts with the two of us. Our network of advisors grew; Luke Nelson and Mike Foote joined as trusted reviewers. In the meantime our business, Uphill Athlete, was growing—more site visits, more questions, more requests for training plans and coaching. When the book deadline was a distant date it was easy to attend to more pressing challenges. As the deadline neared, Scott immersed himself in writing, and the conversations among the three of us became more frequent. We addressed the same questions we have grappled with for years now: How deep to go into physiology? How to teach enough material that people can truly self-coach? How to address complications such as the power needed for a vertical kilometer race and the aerobic capacity needed for the Tor des Géants?

The final result: my climbing-training journey and Scott’s thirty-plus years of athletic and coaching experience, combined with the insights from Kílian and his years in the snow and on the trail. Everything we learned in writing *Training for the New Alpinism*, and all we continue to learn writing training plans, coaching climbers, coaching skimo racers, and coaching mountain runners—it is in your hands. *Training for the Uphill Athlete* is a distillation of hundreds of thousands of hours of racing, training, and coaching experience. Just as *Training for the New Alpinism* was built on applying conventional sport methodologies to the unconventional sport of climbing big mountains, this book too targets athletes in unconventional sports but uses many of the training methods developed and thoroughly tested in the conventional endurance sports.

This book is a carefully selected blend of information aimed at empowering you to make informed decisions using a proven structure of established knowledge.

We know this works because everything in this book is what Scott, Kilian, and I each have lived—from being world-class athletes to becoming world-class coaches and moderators of a top online training resource.

The book also includes stories from other athletes besides Kilian. Do not be tempted to match these athletes' or Kilian's training (volume, technique, or most other aspects of their training). Instead, try to understand the theory behind what they are saying and the workouts they describe and apply the same principles at your own level. Blindly copying the workouts of world-class athletes is a bad idea.

Training for the Uphill Athlete is the anti-fad. It contains proven training knowledge that you can apply to yourself, and get better, every day for decades to come. The only promise is that of smart, hard, consistent work. So please read carefully, make informed training decisions, and get out with your nose to the wind. Enjoy the power of the uphill athlete.

See you in the mountains,

Steve House

Ridgway, Colorado

May 2018

